Forensic License Plate Super-Resolution, Restoration & OCR

Comprehensive Standardized Pipeline Evaluation Report

Executive Performance Summary

- **Total Test Samples Evaluated**: **300 samples** (Denominator N = 300)
- **End-to-End Exact Match Accuracy**: **50.0%**
- **Mean (Macro) Character Error Rate (CER)**: **0.181**
- **Normalized Macro Character Accuracy**: **81.9%**
- **Corpus (Micro) CER**: **0.181**
- **Corpus (Micro) Character Accuracy**: **81.9%**
- **YOLO Detection Recall @ 0.50 IoU**: **99.0%** (True AP@0.50 = 0.987)
- **Mean Localization IoU**: **0.791** (All plates)
- **Average Pipeline Latency**: **3713.0 ms / frame** (~**0.3 FPS** on cuda:0)

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1. License Plate Detection & Localization (YOLOv8)

Metric	Value	Description
Precision @ 0.50 IoU	99.7%	TP / (TP + FP) at IoU >= 0.50
Recall @ 0.50 IoU	99.0%	TP / Total Manifest at IoU >= 0.50
F1-Score @ 0.50 IoU	0.993	Harmonic mean of detection precision and recall
True AP @ 0.50	0.987	Area under Precision-Recall curve at IoU 0.50 (101-point COCO interpolation)
True mAP @ 0.50:0.95	0.534	COCO mAP averaged across IoU thresholds 0.50 to 0.95
Mean IoU (All Plates)	0.791	Average box overlap across all GT plates (misses counted as 0.0)
Mean IoU (Detected Only)	0.796	Average box overlap conditioned on localization
Detection Failure Rate	0.7%	Percentage of images where plate was unlocalized

Detection Stratification by Plate Size:

Scale Bucket	Area Criteria	Test Count	Recall @ 0.50	Mean IoU (All)	Mean IoU (Det)
Medium	32^2 - 96^2 px	167	100.0%	0.781	0.781
Large	>= 96^2 px	99	97.0%	0.836	0.854
Small	< 32^2 px	34	100.0%	0.706	0.706

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2. Geometric Rectification & Deskewing

Metric	Measured Value	Target Standard
Mean Residual Skew Angle	**0.23°**	< 1.0°
Mean Aspect Ratio Error	**19.4%**	< 15.0%

3. Three-Way Baseline Comparison: Progressive Enhancement Impact

Evaluation Mode	Description	Exact Match	Mean CER	Char Accuracy	Corpus CER
Original GT Plate Crop	Original GT Plate Crop -> OCR	**51.7%**	**0.171**	**82.9%**	0.170
Enhanced GT Plate Crop	Enhanced GT Plate Crop -> OCR	**50.3%**	**0.171**	**82.9%**	0.171
Full Autonomous Pipeline	Full Autonomous Pipeline -> OCR	**49.3%**	**0.184**	**81.5%**	0.184

4. OCR Error & Character Confusion Analysis

Top Character Substitution Pairs

Substitution (Actual -> Predicted)	Occurrences	Typical Root Cause
'M' -> 'H'	11	Morphological character geometry similarity
'0' -> 'O'	11	Morphological character geometry similarity
'B' -> '8'	10	Morphological character geometry similarity
'4' -> '2'	9	Morphological character geometry similarity
'1' -> 'I'	6	Morphological character geometry similarity
'M' -> 'N'	6	Morphological character geometry similarity

Alignment Error Breakdown

- **Total Edit Distance (Errors)**: 522 character errors across 2882 ground-truth characters
- **Top Deletions**: 'M' (32x), 'H' (28x), '0' (27x), '1' (26x), '7' (21x)
- **Top Insertions**: '1' (8x), 'E' (4x), 'N' (3x), 'K' (2x), 'F' (2x)

5. Runtime & Throughput Profile

- **Average Frame Latency**: **3713.0 ms**
- **System Throughput**: **0.3 frames / second**

Pipeline Stage	Mean Execution Time (ms)
Deblurring	576.21 ms
Detection	66.24 ms
Geometry	4.80 ms
Qa	10.11 ms
Routing And Restoration	204.21 ms
Ocr And Selection	2851.09 ms
Evaluation	0.00 ms

Report generated autonomously by ForensicPipelineEvaluator.